

## WETASKIWIN AGCM, AB, Canada

WMO: 718950

Lat: 52.9803N

Lon: 113.4442W

Elev: 2539

StdP: 13.40

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p) |
| 12            | -23.4      | -18.1 | -29.1                         | 1.2 | -23.1 | -23.8 | 1.6 | -18.0 | N/A                   | N/A  | N/A | N/A  | N/A                   | N/A  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.9                   | 81.9            | 62.9 | 79.1 | 61.5 | 76.1 | 60.3 | 66.6                | 77.1 | 64.4 | 75.3 | 62.5 | 72.6 | N/A                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.6                            | 93.3 | 72.9 | 60.2 | 85.5 | 69.7 | 58.2 | 79.6 | 67.6 | 32.7          | 77.5 | 30.9 | 75.6 | 29.4 | 72.7 | 77.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) N/A           | N/A  | N/A | DB                         | -32.3 | 88.0               | 5.6 | 3.4                                                | -36.4 | 90.5       | -39.6 | 92.5       | -42.8 | 94.4       | -46.9 | 96.9 | (4) |
| (5)               |      |     | WB                         | -32.5 | 70.6               | 5.6 | 2.5                                                | -36.5 | 72.4       | -39.8 | 73.9       | -43.0 | 75.3       | -47.0 | 77.1 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 37.0   | 14.9  | 12.6  | 23.5  | 37.3  | 50.6  | 57.5  | 61.1  | 59.6  | 51.0  | 38.8  | 23.9  | 11.7  |  |  |
|                                                                        | DBStd       | 21.04  | 14.83 | 14.99 | 13.88 | 9.43  | 7.86  | 5.19  | 4.85  | 5.42  | 7.93  | 8.34  | 12.70 | 13.93 |  |  |
|                                                                        | HDD50       | 5849   | 1087  | 1048  | 820   | 388   | 88    | 7     | 0     | 3     | 81    | 358   | 783   | 1186  |  |  |
|                                                                        | HDD65       | 10250  | 1552  | 1468  | 1285  | 831   | 447   | 231   | 138   | 179   | 422   | 813   | 1233  | 1651  |  |  |
|                                                                        | CDD50       | 1114   | 0     | 0     | 0     | 7     | 108   | 231   | 345   | 301   | 111   | 10    | 1     | 0     |  |  |
|                                                                        | CDD65       | 37     | 0     | 0     | 0     | 0     | 2     | 5     | 17    | 12    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 1094   | 0     | 0     | 0     | 3     | 111   | 131   | 353   | 324   | 167   | 5     | 0     | 0     |  |  |
|                                                                        | CDH80       | 179    | 0     | 0     | 0     | 0     | 14    | 14    | 59    | 56    | 36    | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | PrecAvg     | 18.5   | 0.9   | 0.6   | 0.8   | 1.1   | 2.1   | 3.5   | 3.5   | 2.3   | 1.4   | 0.9   | 0.8   | 0.7   |  |  |
| (15) Precipitation                                                     | PrecMax     | 22.6   | 2.9   | 1.5   | 1.8   | 2.6   | 4.1   | 5.7   | 7.0   | 4.4   | 3.0   | 2.2   | 2.4   | 1.3   |  |  |
|                                                                        | PrecMin     | 13.4   | 0.1   | 0.1   | 0.2   | 0.2   | 0.7   | 1.0   | 1.0   | 0.7   | 0.1   | 0.2   | 0.2   | 0.2   |  |  |
|                                                                        | PrecStd     | 2.5    | 0.6   | 0.3   | 0.4   | 0.6   | 0.9   | 1.1   | 1.4   | 0.8   | 0.8   | 0.5   | 0.5   | 0.4   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 45.4  | 44.5  | 55.0  | 72.4  | 81.9  | 82.1  | 85.5  | 85.3  | 85.0  | 72.4  | 59.0  | 42.9  |  |  |
|                                                                        |             | MCWB   | 37.3  | 37.7  | 42.9  | 50.7  | 55.4  | 62.8  | 65.4  | 64.2  | 60.2  | 53.1  | 46.2  | 36.5  |  |  |
|                                                                        | 2%          | DB     | 42.1  | 39.9  | 48.6  | 65.5  | 77.1  | 77.8  | 81.3  | 80.9  | 79.1  | 64.0  | 49.5  | 39.1  |  |  |
|                                                                        |             | MCWB   | 36.1  | 34.3  | 39.6  | 47.7  | 53.4  | 59.9  | 65.6  | 63.7  | 58.0  | 49.4  | 41.0  | 34.1  |  |  |
|                                                                        | 5%          | DB     | 39.2  | 36.9  | 43.9  | 58.8  | 72.6  | 74.3  | 78.0  | 77.6  | 73.7  | 58.3  | 43.3  | 35.2  |  |  |
|                                                                        |             | MCWB   | 34.8  | 32.6  | 37.2  | 43.9  | 52.4  | 58.1  | 64.1  | 63.0  | 56.7  | 45.8  | 37.1  | 30.9  |  |  |
|                                                                        | 10%         | DB     | 35.5  | 33.6  | 39.7  | 53.8  | 67.9  | 71.0  | 74.6  | 74.2  | 67.9  | 53.4  | 39.6  | 31.3  |  |  |
|                                                                        |             | MCWB   | 31.9  | 30.1  | 34.3  | 41.7  | 50.4  | 56.6  | 62.2  | 61.7  | 54.3  | 43.7  | 34.8  | 28.0  |  |  |
|                                                                        | 0.4%        | WB     | 38.3  | 37.7  | 43.8  | 51.5  | 57.5  | 65.6  | 70.1  | 68.7  | 61.9  | 54.2  | 46.5  | 37.2  |  |  |
|                                                                        |             | MCDB   | 43.5  | 43.7  | 52.7  | 71.0  | 74.9  | 77.6  | 80.8  | 79.2  | 78.0  | 69.9  | 57.8  | 42.2  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 36.4  | 35.0  | 40.2  | 47.7  | 55.3  | 62.3  | 67.4  | 66.4  | 59.7  | 50.4  | 41.5  | 34.2  |  |  |
|                                                                        |             | MCDB   | 41.4  | 39.4  | 47.7  | 64.2  | 71.6  | 73.5  | 77.5  | 77.2  | 75.7  | 62.0  | 48.1  | 38.4  |  |  |
|                                                                        | 5%          | WB     | 34.7  | 32.8  | 37.3  | 45.0  | 53.6  | 60.1  | 65.0  | 64.1  | 57.3  | 47.1  | 37.8  | 31.3  |  |  |
|                                                                        |             | MCDB   | 39.1  | 36.3  | 43.3  | 57.0  | 69.9  | 70.1  | 75.4  | 75.4  | 71.5  | 56.0  | 42.5  | 35.1  |  |  |
|                                                                        | 10%         | WB     | 31.9  | 30.4  | 34.7  | 42.2  | 51.7  | 58.2  | 63.2  | 62.1  | 55.2  | 44.6  | 35.2  | 28.0  |  |  |
|                                                                        |             | MCDB   | 35.2  | 33.9  | 39.0  | 52.3  | 66.0  | 68.0  | 73.1  | 72.2  | 66.6  | 51.9  | 38.8  | 31.1  |  |  |
|                                                                        | 5% DB       | MDBR   | 18.5  | 19.8  | 19.2  | 21.2  | 24.3  | 21.7  | 22.9  | 24.4  | 25.3  | 21.8  | 17.9  | 17.4  |  |  |
|                                                                        |             | MCDBR  | 20.3  | 20.5  | 23.6  | 32.0  | 32.4  | 28.5  | 29.3  | 31.8  | 37.3  | 31.5  | 23.2  | 19.7  |  |  |
|                                                                        |             | MCWBR  | 15.8  | 15.9  | 15.3  | 16.5  | 13.9  | 13.3  | 15.7  | 16.5  | 17.9  | 17.5  | 15.6  | 15.7  |  |  |
|                                                                        |             | MCWBR  | 19.9  | 19.9  | 22.7  | 30.2  | 28.3  | 23.9  | 27.2  | 28.5  | 33.2  | 28.8  | 21.0  | 19.5  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR  | 15.7  | 15.5  | 15.1  | 16.0  | 12.6  | 12.3  | 15.6  | 15.8  | 16.9  | 16.6  | 14.7  | 15.9  |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.259 | 0.264 | 0.302 | 0.324 | 0.340 | 0.347 | 0.350 | 0.360 | 0.308 | 0.282 | 0.253 | 0.243 |  |  |
|                                                                        | taud        |        | 2.308 | 2.383 | 2.372 | 2.388 | 2.394 | 2.391 | 2.421 | 2.354 | 2.487 | 2.538 | 2.462 | 2.312 |  |  |
|                                                                        | Ebn at Noon |        | 235   | 274   | 283   | 289   | 289   | 286   | 283   | 273   | 279   | 266   | 242   | 220   |  |  |
|                                                                        | Edh at Noon |        | 19    | 23    | 30    | 33    | 35    | 36    | 34    | 34    | 26    | 20    | 16    | 15    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 328   | 660   | 1129  | 1491  | 1822  | 1922  | 1990  | 1632  | 1130  | 663   | 328   | 240   |  |  |
|                                                                        | RadStd      |        | 33    | 54    | 84    | 142   | 123   | 123   | 100   | 112   | 119   | 77    | 36    | 23    |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (75 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page